

Chapter 3

Operators and Quantum Dynamics

Chapter 2 introduced wave functions, probability amplitudes, expectation values, uncertainty, and the momentum operator. We now turn that first example of an operator into a general framework for observables and measurements.

Lecture 6: Operators and Observables

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Lecture 5 introduced expectation values, uncertainties, and the momentum operator. This lecture develops the general operator language of quantum mechanics. We define linear operators and eigenfunctions, construct the Hamiltonian, examine the noncommutativity of position and momentum, and connect eigenfunction expansions with measurement outcomes and probabilities.

Questions.

- How does a mathematical operator represent a physical observable?
- What is an eigenfunction, and what does its eigenvalue mean physically?
- Why do position and momentum fail to commute?
- Which values can a measurement return, and how are their probabilities found?
- How does the general measurement rule reproduce the Born rule for position?

3.1 From wave functions to operators

3.1.1 A first example

At a fixed time, the state of a particle is described by a normalized wave function $\psi(x)$. Lecture 5 introduced the momentum operator

$$\hat{p} = -i\hbar \frac{\partial}{\partial x}. \quad (3.1)$$

For a general state, $\hat{p}\psi$ is another function rather than a measured momentum. Plane waves are special because

$$\hat{p}e^{ikx} = \hbar k e^{ikx}. \quad (3.2)$$

The operator returns the same function multiplied by the definite momentum $p = \hbar k$.

3.1.2 Operators as rules

An operator is a rule that maps a function to another function. It is defined by its action on an arbitrary test function $f(x)$. Examples include

$$\hat{1}f(x) = f(x), \quad \hat{x}f(x) = xf(x), \quad (3.3)$$

$$\frac{d}{dx}f(x) = f'(x), \quad \hat{M}_h f(x) = h(x)f(x). \quad (3.4)$$

The domain of an operator specifies which input functions are allowed. This matters for singular operations and for derivatives at boundaries.

3.1.3 Linearity and superposition

Definition 1 (Linear operator). An operator $\hat{O}$ is linear if, for arbitrary complex numbers a, b and functions f, g ,

$$\hat{O}(af + bg) = a\hat{O}f + b\hat{O}g. \quad (3.5)$$

The identity, derivative, and multiplication operators above are linear. The squaring rule $\hat{S}f = f^2$ is not:

$$\hat{S}(f + g) = (f + g)^2 = f^2 + 2fg + g^2 \neq \hat{S}f + \hat{S}g. \quad (3.6)$$

Linearity is required because quantum states obey superposition.

3.2 Eigenfunctions and observables

3.2.1 Eigenfunctions and eigenvalues

Definition 2 (Eigenfunction). A nonzero function $f_a(x)$ is an eigenfunction of $\hat{A}$ with eigenvalue a if

$$\hat{A}f_a(x) = af_a(x). \quad (3.7)$$

The eigenvalue a is a number, while f_a is a function.

The momentum operator provides a familiar example:

$$\hat{p}e^{ikx} = \hbar k e^{ikx}, \quad (3.8)$$

so e^{ikx} is a momentum eigenfunction with eigenvalue $p = \hbar k$.

Multiplying an eigenfunction by a nonzero constant does not change its eigenvalue:

$$\hat{A}(cf_a) = c\hat{A}f_a = a(cf_a). \quad (3.9)$$

By contrast, if

$$\begin{aligned}\hat{A}f_1 &= a_1f_1, & \hat{A}f_2 &= a_2f_2, & a_1 &\neq a_2, \\ \hat{A}(f_1 + f_2) &= a_1f_1 + a_2f_2.\end{aligned}\tag{3.10}$$

The result is not one constant times $f_1 + f_2$, so the sum is generally not an eigenfunction. If $a_1 = a_2$, any linear combination remains an eigenfunction; the eigenvalue is then degenerate.

3.2.2 Position, momentum, and energy operators

Postulate (Operators for observables). Every measurable quantity, or observable, is represented by a linear operator. In the position representation, examples include

$$x \longleftrightarrow \hat{x}, \quad p \longleftrightarrow \hat{p}, \quad E \longleftrightarrow \hat{H}.\tag{3.11}$$

The position operator multiplies a wave function by x , and the momentum operator differentiates it:

$$\hat{x}\psi(x) = x\psi(x), \quad \hat{p}\psi(x) = \frac{\hbar}{i} \frac{\partial \psi}{\partial x}.\tag{3.12}$$

The classical energy of a nonrelativistic particle in a potential $V(x)$ is

$$E = \frac{p^2}{2m} + V(x).\tag{3.13}$$

Replacing position and momentum by their operators gives the energy operator, conventionally called the Hamiltonian:

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x).\tag{3.14}$$

Its action on a wave function is

$$\hat{H}\psi(x) = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x)}{\partial x^2} + V(x)\psi(x).\tag{3.15}$$

Here $V(\hat{x})$ means multiplication by $V(x)$.

Lecture 5's expectation-value rule now takes the general form

$$\langle A \rangle_\psi = \int_{-\infty}^{\infty} \psi^*(x) \hat{A}\psi(x) dx.\tag{3.16}$$

3.3 Commutators and incompatible observables

3.3.1 Operator order

Operator products act from right to left:

$$(\hat{A}\hat{B})f = \hat{A}(\hat{B}f).\tag{3.17}$$

Definition 3 (Commutator). The commutator of $\hat{A}$ and $\hat{B}$ is

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}. \quad (3.18)$$

The operators commute if this is the zero operator.

Because operators are defined by what they do, we determine a commutator by applying it to an arbitrary test function. For position and momentum,

$$\hat{x}\hat{p}f(x) = \frac{\hbar}{i}x\frac{df}{dx}, \quad (3.19)$$

$$\hat{p}\hat{x}f(x) = \frac{\hbar}{i}\frac{d}{dx}(xf(x)) = \frac{\hbar}{i}f(x) + \frac{\hbar}{i}x\frac{df}{dx}. \quad (3.20)$$

Subtracting gives

$$(\hat{x}\hat{p} - \hat{p}\hat{x})f(x) = i\hbar f(x). \quad (3.21)$$

Since this holds for every function in the common domain of the operators,

Postulate (Canonical commutation relation).

$$[\hat{x}, \hat{p}] = i\hbar\hat{1}. \quad (3.22)$$

The nonzero commutator means that position and momentum are incompatible observables. Their uncertainties satisfy

$$\Delta x \Delta p \geq \frac{\hbar}{2}. \quad (3.23)$$

Remark (Why the state space is infinite-dimensional). Suppose that X and P were $N \times N$ matrices. The cyclic property of the trace gives

$$\text{Tr}[X, P] = \text{Tr}(XP) - \text{Tr}(PX) = 0.$$

However, the canonical commutation relation $[X, P] = i\hbar I_N$ would require

$$\text{Tr}[X, P] = i\hbar \text{Tr}(I_N) = i\hbar N \neq 0,$$

which is a contradiction. Thus no pair of finite-dimensional matrices can exactly represent position and momentum. Their operators must act on an infinite-dimensional state space, such as the space of wave functions.

3.4 Measurement in the operator language

3.4.1 Possible outcomes and state reduction

Postulate (Measurement outcomes). When an observable A , represented by $\hat{A}$, is measured, the measured value must be one of the eigenvalues of $\hat{A}$.

Definition 4 (Hermitian operator). An operator $\hat{A}$ is *Hermitian* if, for any states ϕ and ψ in its domain,

$$\langle \phi | \hat{A} \psi \rangle = \langle \hat{A} \phi | \psi \rangle.$$

Equivalently, $\hat{A}^\dagger = \hat{A}$, where $\dagger$ denotes the adjoint (the conjugate transpose for a matrix). For wave functions, this condition reads

$$\int \phi^*(x) (\hat{A} \psi)(x) dx = \int (\hat{A} \phi)^*(x) \psi(x) dx.$$

Hermitian operators have real eigenvalues. If $\hat{A} \psi_a = a \psi_a$ and ψ_a is normalized, Hermiticity gives $a = \langle \psi_a | \hat{A} \psi_a \rangle = \langle \hat{A} \psi_a | \psi_a \rangle = a^*$. Thus a is real. Since measurement results must be real, quantum observables are represented by Hermitian operators.

Postulate (State after an ideal measurement). If measuring A gives a , the state immediately afterward lies in its eigenspace:

$$\hat{A} \psi_a = a \psi_a. \quad (3.24)$$

For nondegenerate a , the state is proportional to ψ_a .

This idealized change is called state reduction or collapse. For example, a momentum result $p = \hbar k$ leaves the system in a state proportional to e^{ikx} .

3.4.2 Definite position and delta functions

Suppose a position measurement returns x_0 . A state perfectly localized at that point is represented formally by

$$\psi_{x_0}(x) \propto \delta(x - x_0). \quad (3.25)$$

This is indeed an eigenfunction of the position operator in the generalized sense:

$$\hat{x} \delta(x - x_0) = x \delta(x - x_0) = x_0 \delta(x - x_0). \quad (3.26)$$

The equality holds because the delta distribution contributes only at $x = x_0$. It is a generalized eigenfunction, not an ordinary normalizable state; a real measurement prepares a narrow packet with finite width.

3.4.3 Eigenfunction expansions and probabilities

Eigenfunctions of a physical observable can be chosen orthonormal:

$$\int_{-\infty}^{\infty} \psi_a^*(x) \psi_b(x) dx = \delta_{ab}. \quad (3.27)$$

For a continuous spectrum, the Kronecker delta is replaced by a Dirac delta.

An arbitrary state can be expanded in this basis. The coefficients give the measurement probabilities:

$$\psi(x) = \sum_a c_a \psi_a(x), \quad P(A = a) = |c_a|^2, \quad (3.28)$$

$$\psi(x) = \int c(a) \psi_a(x) da, \quad P(a \leq A \leq a + da) = |c(a)|^2 da. \quad (3.29)$$

The spectrum supplies the possible outcomes; the state supplies their probabilities.

Position makes this abstract rule concrete. Since $\delta(x - x_0)$ is the generalized eigenfunction with eigenvalue x_0 , the identity

$$\psi(x) = \int_{-\infty}^{\infty} \psi(x_0) \delta(x - x_0) dx_0. \quad (3.30)$$

The coefficient is $\psi(x_0)$, so the position probability density is $|\psi(x_0)|^2$. This is the Born rule.

Questions.

1. Which maps are linear: $\hat{O}f = f'$, $\hat{O}f = xf$, and $\hat{O}f = f^2$?
2. Verify that e^{ikx} is an eigenfunction of $\hat{p} = -i\hbar\partial_x$. What is its eigenvalue?
3. Apply $[\hat{x}, \hat{p}]$ to an arbitrary function $f(x)$ and recover $i\hbar f(x)$.
4. What are the possible outcomes of measuring an observable A ? What determines their probabilities?
5. Why is $\delta(x - x_0)$ a generalized eigenfunction rather than an ordinary physical state?

Lecture 6 summary

- An operator maps functions to functions. Quantum observables are represented by linear, Hermitian operators.
- An eigenfunction satisfies $\hat{A}\psi_a = a\psi_a$; in this state, A has the definite value a .
- In the position representation, $\hat{x}$ multiplies by x , $\hat{p} = -i\hbar\partial_x$, and $\hat{H} = \hat{p}^2/(2m) + V(\hat{x})$.
- Operator order matters. Position and momentum satisfy $[\hat{x}, \hat{p}] = i\hbar\hat{1}$, so they are incompatible observables and obey $\Delta x \Delta p \geq \hbar/2$.
- Measurement outcomes are the eigenvalues of $\hat{A}$. If $\psi = \sum_a c_a \psi_a$, then $P(A = a) = |c_a|^2$.
- Dirac delta distributions are generalized position eigenfunctions and reproduce the Born rule for position.

Suggested reading. Griffiths and Schroeter, *Introduction to Quantum Mechanics*, 3rd ed., Ch. 1, sections on operators, expectation values, and uncertainty. Review Lecture 5 on expectation values, Fourier components, and the momentum operator.